Complexity Theory

TM-Halt — in **RE**

SAT-FO-String — in **R**

Input: A first-order formula φ for strings
Decision: Is φ satisfiable?

2-PlayerCorridorTiling — **EXP**-C

Input: Tilings T and integer k (unary)
Decision: Does player one has a winning strategy in the tiling game?

QBF — **PSPACE**-C

Input: Quantified Boolean formula φ (in prenex normal form and CNF)
Decision: Is φ satisfiable?

CorridorTiling — **PSPACE**-C

Input: Tilings T and integer k (unary)
Decision: Is there a $k \times l$ tiling for T?

SAT-UNSAT — **DP**-C

Input: Boolean formulas φ_1, φ_2 in CNF
Decision: Is φ_1 satisfiable and φ_2 unsatisfiable?

SAT — **NP**-C

Input: Boolean formula φ in CNF
Decision: Is φ satisfiable?

Tiling — **NP**-C

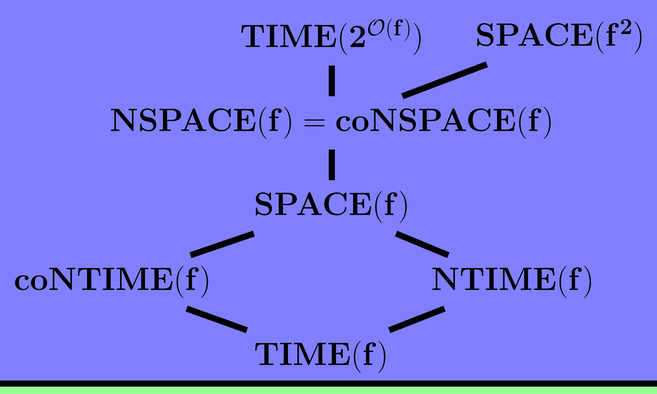
Input: Tilings T und integers c, r (unary)
Decision: Is there a $c \times r$ tiling for T?

TSP — **NP**-C

Input: Weighted Graph G and $k \in \mathbb{N}$
Decision: Is there a circle in G with a weight smaller or equal to k that visits all nodes?

Small Hierarchy

For $f(n) \geq \log(n)$



HornSAT — **P**-C

Input: Finite set of Horn-clauses K
Decision: Is K satisfiable?

Determinant — in $\mathbf{NC^2}$

AcyclicCS — **LOGCFL**-C

Input: Variables V, domain U, constraints C, such that (V, C) is acyclic
Decision: Is there a solution $\sigma \subseteq V \times U$ such that $\sigma(\mathcal{C})$ is true?

REACH — **NL**-C

Input: Graph $G = (V, E)$ and $s, t \in V$
Decision: Is there a path from s to t in G?

UREACH — **L**-C

Input: Undirected graph $G{=}(V, E)$ and $s, t \in V$
Decision: Is there a path from s to t in G?

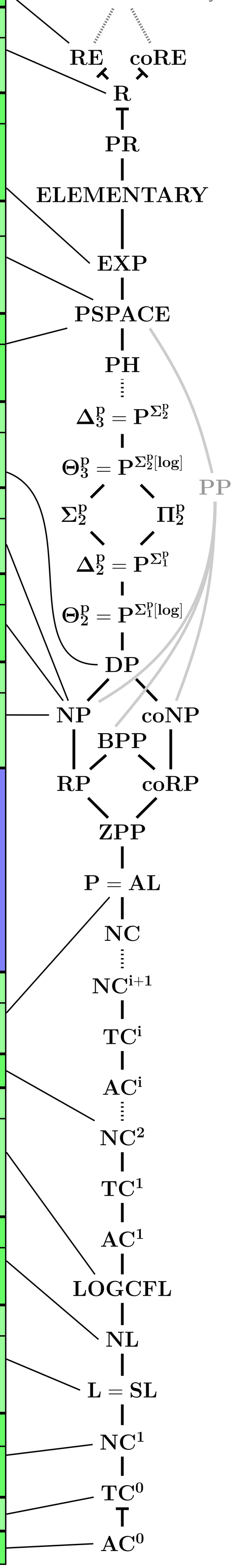
FormulaValueProblem — $\mathbf{NC^1}$-C

Input: Variable free Boolean formula B
Decision: Does B evaluate to true?

Multiplication — $\mathbf{TC^0}$-C

Addition — in $\mathbf{AC^0}$

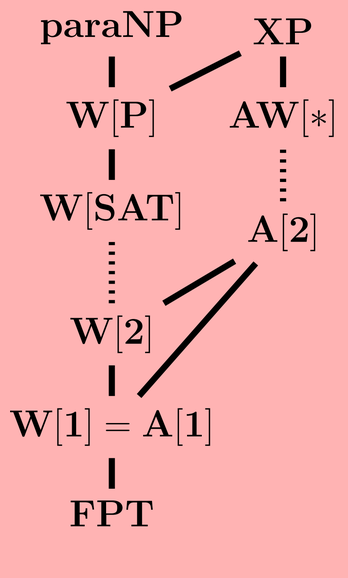
Arithmetical Hierarchy



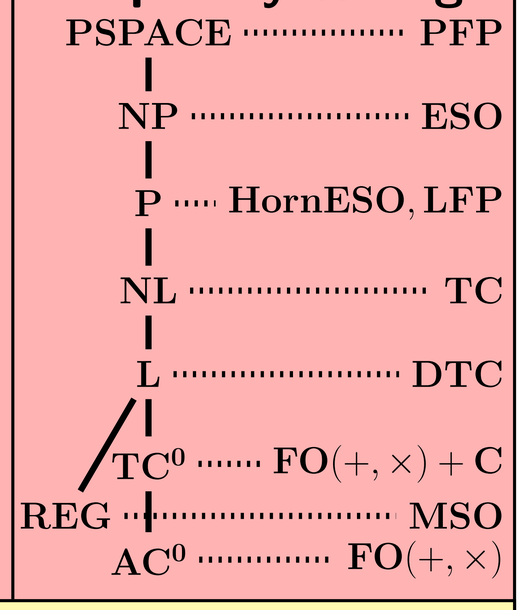
Parameterized Problems

A tuple (Q, κ) is a parameterized problem, if $Q \subseteq \Sigma^*$ and $\kappa : \Sigma^* \to \mathbb{N}$ is computable in polynomial time.

Descriptive Complexity & Parametrization



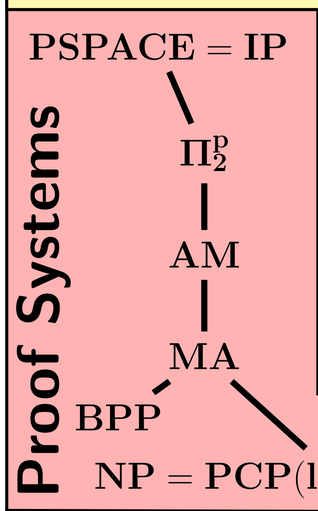
Complexity & Logic



Complexity Classes and Logic

A logic $\mathcal{L}$ has complexity $\mathcal{C}$, if for each sentence $\varphi \in \mathcal{L}$ it holds $\{ \text{enc}(\mathfrak{A}) \mid \mathfrak{A} \models \varphi \} \in \mathcal{C}$.

Proof Systems



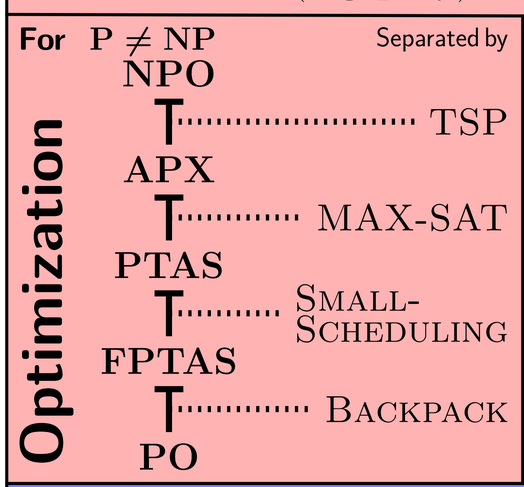
PCP *(probabilistically checkable proof)*

$PCP(\mathcal{F}, \mathcal{G})$ is the class of languages L such there is a verifier V that

- uses $\leq r(n)$ random bits, $(r \in \mathcal{F})$
- reads $\leq q(n)$ bits of the proof, $(q \in \mathcal{G})$
- accepts for a proof π everything from L and does not accept any $x \notin L$ with probability $> \frac{1}{2}$.

$$\mathbf{NP} = \mathbf{PCP}(\log, \text{poly}) = \mathbf{PCP}(0, \text{poly}) = \mathbf{PCP}(\log, 1)$$

Optimization



Class operators

For every $\mathcal{C}$ which is closed under Turing-reduction holds:

$$\begin{array}{rcl} \mathbf{BP} \cdot \mathbf{BP} \cdot \mathcal{C} & \subseteq & \mathbf{BP} \cdot \mathcal{C} \\ \oplus \cdot \oplus \cdot \mathcal{C} & \subseteq & \oplus \mathcal{C} \\ \oplus \cdot \mathbf{BP} \cdot \mathcal{C} & \subseteq & \mathbf{BP} \cdot \oplus \cdot \mathcal{C} \\ \exists \cdot \mathcal{C} & \subseteq & \mathbf{BP} \cdot \oplus \cdot \mathcal{C} \\ \exists \cdot \mathbf{BP} \cdot \mathcal{C} & \subseteq & \mathbf{BP} \cdot \exists \cdot \mathcal{C} \\ \exists \cdot \mathbf{BP} \cdot \mathbf{P} & = & \mathbf{MA} \\ \mathbf{BP} \cdot \exists \cdot \mathbf{P} & = & \mathbf{AM} \end{array}$$

Theorem — Chandra, Kozen, Stockmeyer

(a) For $S(n) = \Omega(\log n)$ holds:
$\mathbf{ASPACE}(\mathbf{S}) \subseteq \mathbf{TIME}(\mathbf{2^{\mathcal{O}(S)}})$

(b) For $T(n) = \Omega(n)$ holds:
$\mathbf{TIME}(\mathbf{T}) \subseteq \mathbf{ASPACE}(\log \mathbf{T})$

Immerman–Szelepcsényi theorem

For space constructible S with $S(n) \geq \log n$ holds:

$$\mathbf{NSPACE}(\mathbf{S}) = \mathbf{coNSPACE}(\mathbf{S})$$

Savitch's theorem — Savitch

For space constructible S with $S(n) \geq \log n$ holds:

$$\mathbf{NSPACE}(\mathbf{S}) \subseteq \mathbf{SPACE}(\mathbf{S^2})$$

Space hierarchy theorem — Stearns, Szepietowski

For time constructible f, g with $f(n) \in \Omega(\log n)$ and $g(n) \in \omega(f(n))$ holds:

$$\mathbf{SPACE}(\mathbf{f}) \subsetneq \mathbf{SPACE}(\mathbf{g})$$
$$\mathbf{NSPACE}(\mathbf{f}) \subsetneq \mathbf{NSPACE}(\mathbf{g})$$

Time hierarchy theorem — Hartmanis, Stearns

For time constructible f, g with $f(n) \in \Omega(n)$ and $g(n) \in \omega(f(n) \cdot \log(f(n)))$ holds:

$$\mathbf{TIME}(\mathbf{f}) \subsetneq \mathbf{TIME}(\mathbf{g})$$

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Based on a lecture by Prof. Schwentick (TU Dortmund University)